

SECTION - A

Question numbers 1 to 6 carry 1 mark each.

1. If vectors $\vec{a}$ and $\vec{b}$ are such that

$$|\vec{a}| = \frac{1}{2}, \quad |\vec{b}| = \frac{4}{5}$$

and

$$|\vec{a} \times \vec{b}| = \frac{1}{\sqrt{3}},$$

then find

$$|\vec{a} \cdot \vec{b}|.$$

2. If $\vec{a}$ and $\vec{b}$ are unit vectors, then what is the angle between $\vec{a}$ and $\vec{b}$ for

$$\vec{a} - \sqrt{2}\vec{b}$$

to be a unit vector?

3. Find the distance between the planes

$$\vec{r} \cdot (2\hat{i} - 3\hat{j} + 6\hat{k}) - 4 = 0$$

and

$$\vec{r} \cdot (6\hat{i} - 9\hat{j} + 18\hat{k}) + 30 = 0.$$

4. If A is a square matrix such that

$$|A| = 5,$$

write the value of

$$|AA'|.$$

5. If

$$A = \begin{bmatrix} 1 & -4 \\ 2 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 3 & 2 \\ 1 & -4 \end{bmatrix},$$

find

$$|AB|.$$

6. If

$$A = \begin{bmatrix} 0 & 3 & 4a \\ 2 & -5 & -8 \end{bmatrix}$$

and

$$kA = \begin{bmatrix} 0 & 3 & 4a \\ 2 & -5 & 5b \end{bmatrix},$$

find the values of k and a .

SECTION - B

Question numbers 7 to 19 carry 4 marks each.

7. Differentiate

$$(\sin 2x)^x + \sin \sqrt{3x}$$

with respect to x .

OR

Differentiate

$$\tan^{-1} \left(\frac{\sqrt{1+x^2} - \sqrt{1-x^2}}{\sqrt{1+x^2} + \sqrt{1-x^2}} \right)$$

with respect to

$$\cos^{-1} x.$$

8. Find k , if

$$f(x) = \begin{cases} k \sin(x+1), & x < 0 \\ \frac{\tan x - \sin x}{x^3}, & x > 0 \end{cases}$$

is continuous at

$$x = 0.$$

9. Find equation of normal to the curve

$$ay^2 = x^3$$

at the point whose x coordinate is

$$am^2.$$

10. Find:

$$\int \frac{1 - \sin x}{\sin x(1 + \sin x)} dx$$

11. Find:

$$\int \left[\log(\log x) + \frac{x}{\log x} \right] dx$$

12. Evaluate:

$$\int_0^{\pi/2} \frac{\sin^{-1} x}{\sin x + \cos x} dx$$

OR

Evaluate:

$$\int_0^1 (1 - x + x^2) dx$$

13. Solve the differential equation:

$$(x + 1) \frac{dy}{dx} - y = e^x (x + 1)^2$$